

Breathing Techniques and Breath-Regulation Strategies in Sport and Performance Contexts: A Scoping Review

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Abstract

Background: Breathing techniques are used in sport as warm-ups, recovery routines, training stimuli, attentional tools, and psychophysiological regulation strategies. Their shared label is misleading because voluntary hyperventilation, hypoventilation at low lung volume, breath holding, slow-paced breathing, diaphragmatic breathing, pranayama, and composite routines manipulate different physiological targets. **Objective:** This scoping review maps non-device breathing and breath-regulation strategies in sport and performance contexts, with attention to technique specificity, outcomes, feasibility, and safety. **Methods:** The review followed PRISMA-ScR principles. Sixty-five reports were charted descriptively. Exploratory syntheses used Hedges' g for extractable efficacy outcomes and risk ratios for extractable dropout and adverse-event data. **Results:** The corpus contained 65 reports published between 2000 and 2026, with 2,188 participants across extractable samples. Crossover and other acute experimental designs were common, while durable training trials were concentrated in selected intervention families. Hypoventilation and breath-holding protocols were most clearly tied to repeated-sprint, swimming, cycling, and apnea contexts; slow-paced and diaphragmatic protocols were more often used for recovery, autonomic regulation, or precision-performance tasks; voluntary hyperventilation was tested mainly as an acute ergogenic or metabolic strategy. Exploratory pooling was uncertain for both performance ($g = -0.06$, 95% CI -0.34 to 0.22) and physiological outcomes ($g = 0.28$, 95% CI -0.07 to 0.63). Dropout and adverse-event syntheses did not show clear comparative differences, but reporting was sparse. **Conclusions:** The evidence does not support a single conclusion about "breathwork" in sport. It supports a technique-specific research agenda built around clear protocols, mechanism-matched outcomes, credible comparators, adherence reporting, and prespecified safety monitoring.

Key takeaways

- The review includes 65 reports, but the evidence is better understood by technique family than by a generic breathwork label.
- Hypoventilation and breath-holding studies provide the clearest sport-training cluster, especially for repeated-sprint and swimming contexts.
- Slow-paced, box, coherent, and diaphragmatic routines are more plausibly positioned as recovery, autonomic, attentional, or pre-performance tools than as universal ergogenic aids.
- Exploratory pooled effects are uncertain and should guide study design rather than practice claims.
- Feasibility and adverse-event reporting remain too incomplete for strong safety reassurance.

Keywords: scoping review; PRISMA-ScR; breathing training; breathwork; hyperventilation; hypoventilation; paced breathing; sport performance; feasibility; safety

1 Introduction

Breathing is both automatic and voluntarily adjustable. Athletes can alter respiratory frequency, depth, inhalation-exhalation ratio, route of breathing, breath-hold timing, and lung volume without external equipment. That makes breath regulation attractive in sport: it can be used before competition, during exertion, between repeated efforts, during recovery, or as a repeated training exposure. It is also a source of conceptual confusion. A pre-sprint hyperventilation protocol, a low-lung-volume hypoventilation session, a box-breathing recovery routine, a pranayama program, and a diaphragmatic warm-up do not share the same mechanism simply because all involve deliberate breathing.

The sport literature already points toward this distinction. Laborde and colleagues separated slow-paced breathing, fast-paced breathing, voluntary hyperventilation, breath holding, and nostril-breathing approaches in their systematic review and meta-analysis, and concluded that intervention duration, technique type, heterogeneity, and risk of bias limited broad claims about physical sport performance (Laborde et al., 2024). The present review takes a complementary scoping approach. Instead of asking whether breathing techniques have a single average effect, it asks what has been studied, where the evidence clusters, which outcomes have been used, and whether feasibility and safety are reported well enough to support translation.

This framing is important because the plausible target changes by technique. Voluntary hyperventilation can alter carbon dioxide and acid-base balance before or between high-intensity efforts. Hypoventilation and end-expiratory breath holding are often used to create hypoxic

or hypercapnic stress during repeated-sprint work. Slow-paced, coherent, box, and diaphragmatic breathing are more often justified through autonomic regulation, attentional control, and recovery. Glossopharyngeal, hook-breathing, and apnea-related routines belong to still different sport-specific practices. A scoping review is therefore appropriate because it can map these families without forcing them into a single efficacy estimate.

The review questions were: (1) which breathing-technique families have been studied in sport and performance contexts; (2) which populations, designs, outcomes, and intervention doses characterize the evidence base; and (3) what can be learned from extractable efficacy, feasibility, and safety data while preserving technique specificity.

2 Methods

2.1 Design and reporting

This scoping review was designed to map the breadth, structure, and reporting completeness of evidence on breathing techniques in sport and performance contexts. The manuscript is reported according to PRISMA-ScR principles. The review maps the available evidence rather than claiming a single definitive intervention effect. The unit of descriptive synthesis was the individual report. Quantitative syntheses were exploratory adjuncts to the evidence map and used only outcomes for which the required group-level data could be extracted.

2.2 Protocol and registration

The review protocol was registered on the Open Science Framework: https://osf.io/3t2wc/overview?view_only=7518720a5f334682b8bb5ebba89b1cbd.

2.3 Eligibility criteria

Eligibility was defined using a Population–Concept–Context structure. Eligible populations included athletes, sport participants, physically active participants, and healthy participants completing performance-relevant exercise or psychophysiological tasks. The concept was a breathing technique, breath-regulation routine, or breathing-related exposure that could be performed without a device that directly altered respiratory load or inspired gas composition. Eligible concepts included hyperventilation, hypoventilation, paced or slow breathing, breath holding/apnea, pranayama or yogic breathing, diaphragmatic breathing, hook breathing, Valsalva-type strategies, and related approaches. Eligible contexts included sport training, laboratory exercise tasks, sport-specific testing, recovery, and performance-relevant psychophysiological outcomes. Source types included primary empirical reports with extractable study-level information; reviews were used for citation searching but were not included as primary evidence rows. The operational inclusion and exclusion criteria are reproduced in the appendix.

2.4 Information sources and search strategy

Searches were run in PubMed, Scopus, and SPORTDiscus. The database-specific search strings are reported in the appendix. The searches combined terms for healthy or athletic populations, breathing techniques, sport or performance outcomes, and exclusions for reviews, respiratory muscle training, pulmonary disease, and major clinical respiratory contexts. Additional reports were identified through citation searching, backtracking from relevant reviews, and manual reconciliation during full-text assessment.

2.5 Selection process

Records were screened at title/abstract level, followed by full-text assessment of potentially eligible reports. Full texts were assessed against the population, concept, context, and source-type criteria described above. Disagreements and uncertain cases were resolved during eligibility reconciliation before final extraction.

2.6 Data charting and quality control

Data were charted for bibliometry, study design, sport context, athlete level, sample size, age, sex distribution, baseline characteristics, breathing category, intervention dose, comparator, efficacy outcomes, uncertainty, feasibility, dropout, adverse events, and safety. Missing or unreported fields were retained as not reported so that denominators and reporting gaps remained explicit. Data-quality checks flagged incomplete efficacy fields, non-positive dispersion values, unclear breathing categories, and sparse feasibility or safety reporting for manual review.

2.7 Outcome grouping and synthesis

Descriptive synthesis summarized publication year, study design, participant characteristics, sport context, breathing category, intervention duration, intervention frequency, feasibility, and safety. Outcomes were grouped into performance, physiological, feasibility, and safety domains. Exploratory meta-analysis used Hedges' g for efficacy outcomes when group-level data were extractable. Effect direction was standardized so that positive values favored the breathing intervention; lower-is-better outcomes such as time, reaction time, heart rate, RPE, dyspnea, fatigue, lactate, and decrement scores were reversed. Safety and dropout were synthesized as risk ratios where event data were available. Very imprecise safety estimates were retained in extraction records but excluded from displayed forest plots when the interval made the figure unreadable.

2.8 Feasibility and safety definitions

Feasibility fields included adherence, dropout, acceptability when available, and reasons for discontinuation. Safety fields included adverse-event reporting status, adverse-event descriptions,

and event counts when extractable. Absence of reported adverse events was not interpreted as evidence that no events occurred unless the source explicitly stated this.

2.9 Critical appraisal

Formal risk-of-bias appraisal was not the primary objective of this scoping review. Instead, the synthesis emphasizes reporting completeness, design heterogeneity, missing feasibility fields, and uncertainty around quantitative estimates. This approach is appropriate for mapping a heterogeneous field, but it limits causal inference and prevents strong claims about efficacy.

3 Results

3.1 Study selection and corpus

Table 1: Aggregate study and participant characteristics.

Characteristic	Summary
Included reports	65 reports
Publication years	2000–2026
Reports contributing total sample size	65 reports
Aggregate reported sample size	2,188 participants
Age reporting	35 reports; weighted mean 26.6 years; pooled SD 9.5 years
Sex reporting	1,538 participants with sex data; estimated male proportion 60.0%
Athlete level	Competitive: 33; physically active: 14; elite: 6; recreationally trained: 3; healthy non-athlete: 2; not reported/blank: 7
Design profile	Experimental: 60 (92.3%); observational: 5 (7.7%)

The included evidence base comprised 65 reports published from 2000 to 2026, with an aggregate extractable sample of 2,188 participants. The corpus was not a coherent set of comparable trials. It combined observational physiology reports such as Durand et al.’s work on exercise-induced arterial hypoxemia (Durand et al., 2000), crossover experiments such as Sakamoto et al.’s hyperventilation studies (Sakamoto et al., 2014, 2015, 2018; Sakamoto et al., 2020), and training trials such as Woorons et al.’s voluntary-hypoventilation programs (Woorons et al., 2024; Woorons, Millet, & Mucci, 2019; Woorons et al., 2008, 2016, 2020b). Table 1 provides the aggregate description; the full study-level extraction is reported in Supplementary Table S1.

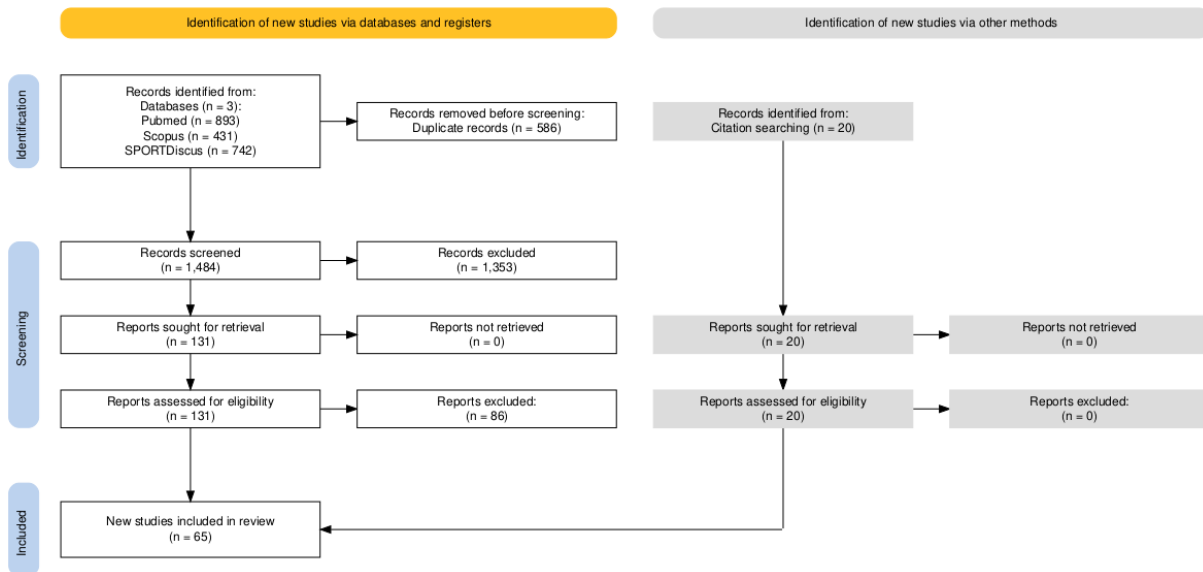


Figure 1: PRISMA flow diagram for study identification, screening, eligibility assessment, and inclusion.

The selection process identified 65 included reports after database screening, full-text assessment, citation searching, and manual reconciliation. Figure 1 is retained in the main manuscript because it is the only figure required to understand how the corpus was assembled. Distribution plots and forest plots are placed in Supplementary Figures S1–S12; these figures document the map but are not needed to follow the main narrative.

3.2 How the evidence base is structured

Table 2: Evidence-map summary.

Domain	Main result
Detailed study designs	Crossover: 33 (50.8%); parallel or controlled trial: 24 (36.9%); single-group/observational: 5 (7.7%); before-after: 3 (4.6%)
Most represented sport contexts	Cycling: 12 (18.5%); swimming: 11 (16.9%); non-sport exercise/performance: 9 (13.8%); running: 9 (13.8%); team sports: 8 (12.3%); resistance/strength: 5 (7.7%)
Breathing categories	Paced breathing: 15 (23.1%); hypoventilation: 14 (21.5%); hyperventilation: 12 (18.5%); breath holding/apnea: 10 (15.4%); other/unclear: 7 (10.8%); pranayama/yogic: 4 (6.2%); diaphragmatic: 3 (4.6%)
Dose duration	26 reports with extractable duration; median 28 days; IQR approximately 0.005–33.8 days
Session frequency	Session frequency was not reported consistently enough for quantitative summary

Table 2 summarizes the map. Crossover designs formed the largest detailed design group, and this matters because acute within-participant experiments answer different questions from repeated training trials. Kacoglu et al. tested brief hyperventilation before jump and sprint tasks (Kaşoğlu & Işık, 2017); Gray et al. tested short pre-race breathing routines in swimming (Gray et al., 2018); Kasap and Aydin compared box breathing and six-breaths-per-minute breathing during post-HIIT recovery (Kasap & Aydin, 2025). These papers are useful for acute strategy questions, but they do not provide the same evidence as repeated hypoventilation training trials in swimmers, rugby players, basketball players, ice-hockey players, soccer players, and judo athletes (Ait Ali Braham et al., 2024; Brocherie et al., 2023; Fornasier-Santos et al., 2018b; Lapointe et al., 2020b; Lőrinczi et al., 2025b; Trincat et al., 2017b; Woorons et al., 2016).

The sport distribution also shaped interpretation. Cycling studies included acute hyperventilation or breath-holding physiology (Dobashi et al., 2021; Fujii et al., 2015; Grossman et al., 2021) and hypoventilation training transfer questions (Woorons, Millet, & Mucci, 2019; Woorons et al., 2020b). Swimming studies included apnea, controlled-frequency breathing, pranayama, and hypoventilation trials (Burtch et al., 2017b; Guimard et al., 2014; Hakkeed et al., 2017; Robertson et al., 2020; Stavrou et al., 2015; Trincat et al., 2017b). Strength or resistance contexts were narrower: Sakamoto et al. tested hyperventilation-aided recovery during bench press and leg press (Sakamoto et al., 2020), Deniz and Erdemir compared breathing strategies during bench press (Deniz & Erdemir, 2025), and Katz et al. tested a diaphragmatic breathing warm-up before deadlift (Katz et al., 2026).

3.3 Voluntary hyperventilation

Hyperventilation was used mainly as an acute strategy. Sakamoto et al. reported that pre-exercise hyperventilation improved repeated cycling sprint performance in one protocol (Sakamoto et al., 2014), but the same group found that hyperventilation-induced alkalosis did not prevent fatigue in maximal voluntary contraction testing (Sakamoto et al., 2015). Their later cycling experiment suggested that hyperventilation duration matters: 15 seconds and 45 seconds during recovery did not reproduce the expected ergogenic effect, whereas the authors interpreted approximately 30 seconds as a more plausible target for that exercise type (Sakamoto et al., 2018). In resistance exercise, Sakamoto et al. reported more repetitions and better maintenance of lifting velocity after hyperventilation-aided recovery than after spontaneous breathing (Sakamoto et al., 2020).

Other hyperventilation studies were less supportive or targeted different outcomes. Fujii et al. and Dobashi et al. showed that hypocapnic hyperventilation shifted metabolic responses during Wingate exercise without clearly improving exercise performance (Dobashi et al., 2021; Fujii et al., 2015). Jacob et al. reported improved 50-m front-crawl performance after pre-exercise hyperventilation (Jacob et al., 2015), whereas Fessler et al. did not support pre-exercise hyperventilation for inclined sprint-interval performance (Fessler et al., 2026). Malakhov et al. found worse balance metrics after hyperventilation relative to quiet breathing (Malakhov et al.,

2014), which is a reminder that an arousal or acid-base manipulation may be undesirable when postural control is central.

3.4 Hypoventilation and end-expiratory breath holding

The clearest sport-training cluster concerned voluntary hypoventilation, often at low lung volume. Woorons et al. tested four weeks of low-pulmonary-volume voluntary hypoventilation in runners (Woorons et al., 2008), five weeks of supramaximal hypoventilation training in swimmers (Woorons et al., 2016), and repeated-sprint cycling or running protocols in later work (Woorons, Dupuy, et al., 2019; Woorons, Millet, & Mucci, 2019; Woorons et al., 2017, 2020b). Trincat et al. found that hypoventilation-induced hypoxia during swimming repeated-sprint training increased the number of sprints completed in a post-training test (Trincat et al., 2017b). Fornasier-Santos et al. reported improved running repeated-sprint ability in rugby players after repeated-sprint training with voluntary hypoventilation (Fornasier-Santos et al., 2018b). Lapointe et al. reported basketball-specific repeated-sprint changes alongside muscle oxygenation, myoelectrical, and potassium-related measures (Lapointe et al., 2020b).

More recent reports extended this line into team and combat sports. Brocherie et al. reported improved off-ice repeated-sprint ability in youth ice-hockey players after off-season RSH-VHL (Brocherie et al., 2023). Ait Ali Braham et al. compared low- and high-lung-volume hypoventilation strategies in female soccer players and reported gains in repeated-sprint ability, with stronger hypoxemia under the low-lung-volume condition (Ait Ali Braham et al., 2024). Woorons et al. reported that elite judo athletes improved the second half of a repeated-sprint rowing test after four weeks of RSH-VHL with maximal end-expiratory breath holding (Woorons et al., 2024). Lorinczi et al. reported additional repeated-sprint benefits in semiprofessional soccer players when prolonged end-expiratory breath holding was added to repeated-sprint training (Lorinczi et al., 2025b). This cluster is therefore promising, but it is also the cluster where oxygen saturation, breath-hold timing, supervision, and stopping rules are most essential to report.

3.5 Breath holding, apnea, and freediving-related routines

Breath-holding and apnea studies were not interchangeable with hypoventilation training. Guimard et al. examined acute apnea swimming responses (Guimard et al., 2014); Lim et al. and Grossman et al. used periodic breath holding during heavy-intensity cycling or Fartlek-style exercise (Grossman et al., 2021; Lim et al., 2018); Bouten et al. reported that acute apnea did not improve 3-km cycling time-trial performance (Bouten, Colosio, Bourgois, Lootens, Van Eenoo, et al., 2020); Robertson et al. tested pre-race apneas before 400-m freestyle swimming (Robertson et al., 2020). These papers show why timing matters: apnea can be used as a pre-performance stimulus, an exercise constraint, or a recovery-related maneuver, but those uses impose different physiological demands.

Freediving and hook-breathing reports contributed a separate safety-relevant thread. Lindholm and Nyrén examined glossopharyngeal breathing in breath-hold divers (Lindholm & Nyrén, 2005). Fernandez et al. reported that hook breathing facilitated oxygen-saturation recovery after deep dives in freedivers with slow recovery (Fernandez et al., 2019). DeAsis-Fernandez et al. reported faster oxygen-saturation recovery after a -40 m apnea dive with the hook maneuver, while ultrasound B-lines and pulmonary-edema-related events were recorded in the safety extraction (DeAsis-Fernandez et al., 2025). The apnea-specific papers therefore should not be used as generic evidence for breathing interventions in land-based sport; they are best read as sport-specific evidence where hypoxemia, pressure, and recovery protocols are central.

3.6 Slow-paced, box, coherent, and diaphragmatic breathing

Slow-paced and related breathing routines were usually framed around autonomic regulation, recovery, stress response, or precision performance. Laborde et al. found that slow-paced breathing after physical exertion improved inhibition outcomes and produced a small favorable effect for burpee performance in the extractable data (Laborde et al., 2019). You et al. showed that a brief slow-paced breathing session increased vagally mediated heart-rate-variability indices in athletes (You et al., 2024). Iskra et al. compared slow-paced breathing at 6 cycles per minute with faster paced breathing at 15 cycles per minute and reported different autonomic patterns (Iskra et al., 2026).

The applied sport findings were mixed because the target tasks differed. Pelka et al. studied slow-paced breathing as a recovery-oriented psychological relaxation technique (Pelka et al., 2017). Conlon et al. found that a five-minute slow diaphragmatic breathing intervention did not clearly improve shooting accuracy relative to a 12-cycles-per-minute paced-breathing comparator, although RMSSD was extractable as a physiological outcome (Conlon et al., 2022a). Buxton et al. tested 4-7-8 slow breathing during inter-set recovery in back squat performance (J. Buxton et al., 2024). Kasap and Aydin compared box breathing with six-breaths-per-minute breathing after HIIT and found physiological recovery differences that depended on the breathing pattern (Kasap & Aydin, 2025). Chiron et al. tested coherent breathing in elite runners during an eight-day intervention (Chiron et al., 2026). Jones et al. reported that box breathing and cyclic sighing accelerated parasympathetic reactivation after maximal exertion in tactical athletes (Jones et al., 2026b). Raidl et al., by contrast, reported no evidence that slow-paced breathing improved recovery after sprint-interval training (Raidl et al., 2026).

Diaphragmatic breathing was narrower but practically relevant. Sikora et al. observed that diaphragmatic breathing patterns in endurance-trained athletes were associated with better pulmonary-function measures than thoracic patterns (Sikora et al., 2024). Katz et al. reported a small acute increase in deadlift 1RM after a five-minute diaphragmatic breathing warm-up (Katz et al., 2026). These findings justify targeted trials, but observational breathing-pattern

associations and acute lifting effects should not be merged into one claim.

3.7 Pranayama, yogic, Wim Hof-type, and composite routines

Composite routines were harder to interpret because breathing was often combined with traditions, meditation, cold exposure, posture, or breath holds. Stavrou et al. and Hakked et al. reported improved pulmonary-function outcomes in young competitive swimmers after yogic or pranayama-style breathing programs (Hakked et al., 2017; Stavrou et al., 2015). Vaidya et al. tested yoga in cricket bowlers (Vaidya et al., 2021), and Bahenský et al. reported altered ventilatory patterns after a yoga-based breathing intervention in young endurance athletes (Bahensk½ et al., 2021). Devipriya et al. reported immediate reaction-time improvements after Bhramari pranayama in young adults (T et al., 2026).

Wim Hof-type interventions illustrate the same interpretive problem. Citherlet et al. tested an acute Wim Hof breathing method before repeated sprinting (Citherlet et al., 2021); Marko et al. found that four weeks of Wim Hof breathing did not clearly improve breathing economy during exercise in adolescent runners (Marko et al., 2022); Fox et al. tested a semi-randomized Wim Hof method intervention that included more than isolated breathing practice (Fox et al., 2025). These reports are valuable, but their mechanisms cannot be assigned to breathing alone unless protocols isolate the active components.

3.8 Exploratory efficacy synthesis

Table 3: Exploratory quantitative syntheses.

Synthesis	k	Estimate	95% CI	I^2	Interpretation
Performance efficacy, Hedges' g	21	-0.06	-0.34 to 0.22	74.8%	Small pooled direction near the null, with confidence interval crossing zero and a wide prediction interval.
Physiological efficacy, Hedges' g	36	0.28	-0.07 to 0.63	89.5%	Positive pooled direction, but uncertainty and heterogeneity are substantial.
Adverse events, risk ratio	3	1.00	0.37 to 2.72	0.0%	No clear overall difference in reported adverse-event risk; event data remain sparse.
Dropout, risk ratio	8	0.82	0.37 to 1.81	0.0%	No clear difference in dropout risk between breathing and comparator conditions.

Table 3 reports the exploratory quantitative syntheses. The pooled performance estimate was close to zero and uncertain. This average is unsurprising once the source studies are read by technique: Malakhov et al.'s balance outcomes after hyperventilation favored quiet breathing (Malakhov et al., 2014), Burtch et al.'s controlled-frequency breathing trial reduced inspiratory muscle fatigue without improving swimming performance (Burtch et al., 2017b), Woorons et al.'s RSH-VHL training studies produced favorable repeated-sprint signals (Woorons et al., 2024; Woorons, Millet, & Mucci, 2019), and Katz et al. found a small deadlift effect after diaphragmatic breathing (Katz et al., 2026). A single pooled performance estimate therefore conceals more than it explains.

The physiological synthesis was directionally positive but highly heterogeneous. The most favorable extractable physiological effects included pulmonary-function outcomes in Stavrou et al. and Hakked et al. (Hakked et al., 2017; Stavrou et al., 2015), HRV outcomes in You et al. and Iskra et al. (Iskra et al., 2026; You et al., 2024), and pulmonary-function differences by breathing pattern in Sikora et al. (Sikora et al., 2024). Physiological outcomes were mechanistically closer to breathing manipulation than performance outcomes, but they spanned oxygenation, ventilation, pulmonary function, HRV, heart rate, blood pressure, and perceptual variables; they should not be treated as one construct.

3.9 Feasibility and safety

Table 4: Feasibility and safety reporting.

Field	Summary
Adherence	One report had an extractable adherence value. Adherence was too sparsely and inconsistently reported for meaningful quantitative summary.
Dropout	Fifty-eight reports provided extractable dropout information, although most recorded no dropout; six reports had non-zero dropout and eight studies contributed to the comparative dropout risk-ratio synthesis.
Adverse-event mentions	Extractable adverse-event descriptions were sparse; one pulmonary-edema mention was counted in the descriptive adverse-event table.
Displayed safety forest plot	Three studies contributed to the comparative adverse-event risk-ratio synthesis. No study was excluded from the displayed safety forest plot.

Table 4 shows that feasibility and safety reporting lagged behind efficacy reporting. Hakked et al. reported 100% adherence for pranayama/yogic breathing (Hakked et al., 2017), but extractable adherence was rare. Dropout or exclusion information was more common but often reflected technical issues, unrelated injuries, or study logistics rather than intervention tolerability: Burtch et al. cited travel, illness, and non-compliance with training sessions (Burtch et al., 2017b); Sakamoto et al. excluded participants because of transducer error (Sakamoto et al., 2018); Iskra et al. and Jones et al. reported technical exclusions affecting physiological datasets (Iskra et al., 2026; Jones et al., 2026b).

Adverse-event reporting was too sparse to support strong reassurance. Sakamoto et al. explicitly reported no hyperventilation-related discomfort in the resistance-exercise recovery study (Sakamoto et al., 2020), and Tomita et al. reported completion without adverse events in the focused-attention/controlled-breathing protocol (Tomita et al., 2026). By contrast, DeAsis-Fernandez et al. reported pulmonary-edema-related findings in a freediving hook-maneuver context (DeAsis-Fernandez et al., 2025). This contrast should guide future reporting: land-based slow-breathing recovery studies, hypoventilation repeated-sprint studies, and deep-apnea studies need different adverse-event definitions and monitoring thresholds.

4 Discussion

4.1 Principal findings

This review maps a field with enough activity to justify technique-specific research, but not enough consistency to justify a single practice recommendation. The strongest narrative cluster is not generic breathwork; it is voluntary hypoventilation or end-expiratory breath holding

embedded in repeated-sprint training, with favorable signals reported by Trincat et al., Fornasier-Santos et al., Lapointe et al., Brocherie et al., Ait Ali Braham et al., Woorons et al., and Lorinczi et al. (Ait Ali Braham et al., 2024; Brocherie et al., 2023; Fornasier-Santos et al., 2018b; Lapointe et al., 2020b; Lőrinczi et al., 2025b; Trincat et al., 2017b; Woorons et al., 2024). The slow-paced and diaphragmatic literature is better interpreted as regulation and recovery evidence, with HRV and stress-response outcomes in You et al., Iskra et al., Chiron et al., Jones et al., and Kasap and Aydin (Chiron et al., 2026; Iskra et al., 2026; Jones et al., 2026b; Kasap & Aydin, 2025; You et al., 2024). Hyperventilation evidence is more conditional: Sakamoto et al. reported benefits in selected repeated-sprint or resistance-recovery contexts (Sakamoto et al., 2014; Sakamoto et al., 2020), but Fujii et al., Dobashi et al., Sakamoto et al., and Fessler et al. show that metabolic shifts do not automatically translate into performance improvement (Dobashi et al., 2021; Fessler et al., 2026; Fujii et al., 2015; Sakamoto et al., 2018).

4.2 Comparison with previous literature

The findings align with Laborde et al.'s conclusion that breathing techniques in sport should be separated by technique, dose, and context rather than pooled into a broad claim (Laborde et al., 2024). The present review extends that conclusion by showing where the map is dense enough for targeted trials and where it remains descriptive. Laborde et al. focused on physical sport performance; this review also charts feasibility, dropout, adverse events, intervention timing, sport context, and physiological outcomes. That broader lens explains why a near-null pooled performance estimate can coexist with plausible technique-specific signals in repeated-sprint hypoventilation, autonomic recovery, pulmonary-function outcomes, and selected strength tasks.

The slow-breathing findings are also consistent with mechanistic literature outside sport. Zaccaro et al. reviewed slow breathing below 10 breaths per minute and described autonomic, central nervous system, and psychological changes, including increased HRV and respiratory sinus arrhythmia, EEG changes, and reduced arousal-related symptoms in healthy participants (Zaccaro et al., 2018). That framework supports interpreting You et al., Iskra et al., Jones et al., Chiron et al., and Kasap and Aydin as regulation and recovery studies rather than direct performance-enhancement trials (Chiron et al., 2026; Iskra et al., 2026; Jones et al., 2026b; Kasap & Aydin, 2025; You et al., 2024). Fincham et al.'s meta-analysis of randomized breathwork trials reported lower self-reported stress, anxiety, and depression after breathwork than control conditions, while also emphasizing heterogeneity, risk-of-bias concerns, and limited adverse-event reporting (Fincham et al., 2023). The parallel with the present review is direct: breathing practices may be useful, but translation requires precise intervention descriptions, credible controls, and safety reporting.

4.3 Implications for practice

The practical message is specificity. For repeated-sprint training in sports such as swimming, rugby, basketball, ice hockey, soccer, and judo, the most relevant evidence comes from hypoventilation and end-expiratory breath-holding protocols (Ait Ali Braham et al., 2024; Brocherie et al., 2023; Fornasier-Santos et al., 2018b; Lapointe et al., 2020b; Lőrinczi et al., 2025b; Trincat et al., 2017b; Woorons et al., 2024). These protocols should be supervised and should report oxygen saturation, symptoms, and stopping criteria. For post-exertion autonomic recovery, the more relevant evidence comes from slow-paced, box, coherent, cyclic-sighing, or diaphragmatic routines (Chiron et al., 2026; Jones et al., 2026b; Kasap & Aydin, 2025; You et al., 2024). For acute ergogenic use of hyperventilation, practitioners should be cautious because benefits appear task- and timing-dependent, and balance or precision demands may make hyperventilation counterproductive (Fessler et al., 2026; Malakhov et al., 2014; Sakamoto et al., 2014; Sakamoto et al., 2020).

4.4 Implications for research

Future trials should stop asking whether breathwork works and instead test named techniques against mechanism-matched outcomes. A hypoventilation training trial should specify lung volume, breath-hold timing, sprint structure, oxygen saturation, supervision, and stopping criteria. A slow-breathing recovery trial should specify respiratory frequency, inhalation-exhalation ratio, session timing, use of cues or biofeedback, and whether the primary target is HRV, perceived recovery, stress, or subsequent performance. A hyperventilation trial should specify duration, timing relative to exertion, target ventilation, symptoms, and whether the expected mechanism is acid-base change, arousal, or recovery between efforts. Composite interventions such as yoga-based breathing and Wim Hof-type methods should either isolate the breathing component or explicitly state that the intervention is multicomponent (Bahensk¹/₂ et al., 2021; Citherlet et al., 2021; Fox et al., 2025; Marko et al., 2022).

Feasibility and safety should be treated as outcomes, not annotations. The next generation of studies should report adherence, acceptability, dropout reasons, intervention fidelity, adverse-event definitions, symptom checklists, and event denominators. This is especially important for interventions that deliberately alter carbon dioxide, oxygen saturation, intrathoracic pressure, or consciousness-related sensations.

4.5 Limitations

This review did not conduct formal risk-of-bias appraisal. That is consistent with its scoping purpose, but it limits causal interpretation. The eligibility criteria were broad enough to include acute crossover experiments, training trials, observational physiology studies, and composite interventions, which increased heterogeneity. Several extraction fields were incompletely reported

in the source papers, especially adherence, intervention fidelity, symptoms, and adverse-event monitoring. The exploratory Hedges' g and risk-ratio syntheses depend on extractable group-level data and should be read as adjuncts to the map, not as definitive estimates of clinical or sport-practice effectiveness.

5 Conclusions

Breathing techniques in sport are not one intervention. The mapped evidence supports technique-specific research: repeated-sprint hypoventilation and end-expiratory breath holding deserve targeted sport-training trials; slow-paced, box, coherent, and diaphragmatic breathing deserve mechanism-matched recovery and regulation trials; hyperventilation requires careful task selection and safety monitoring; composite routines require clearer component reporting. Current pooled estimates are uncertain, and feasibility and safety reporting are too sparse for strong implementation claims. Future research should prioritize precise protocols, credible comparators, complete adherence and adverse-event reporting, and outcomes matched to the proposed mechanism.

Declarations

Funding: No external funding was declared.

Conflicts of interest: The authors declare no conflicts of interest.

Data availability: The extraction data, generated tables, plots, and scripts are available in the project repository.

Author contributions: The authors contributed to study conception, data charting, analysis, interpretation, and manuscript preparation.

A PRISMA-ScR Checklist Crosswalk

Table 5: PRISMA-ScR checklist crosswalk.

Item	Reporting location	Status
Title and abstract	Title and structured abstract identify the manuscript as a scoping review and summarize source selection, charting, and synthesis.	Reported
Rationale	Introduction explains why heterogeneous breathing strategies require mapping before definitive efficacy recommendations.	Reported

Item	Reporting location	Status
Objectives	Introduction states mapping objectives for study characteristics, interventions, outcomes, feasibility, safety, and exploratory effects.	Reported
Eligibility criteria	Methods describe eligible breathing techniques, populations, and sport/performance contexts.	Reported
Information sources	Methods identify the databases and supplementary citation-searching methods.	Reported
Selection of sources	Results include Figure 1.	Reported
Data charting	Methods describe bibliometric, participant, intervention, outcome, feasibility, and safety fields.	Reported
Synthesis of results	Results combine descriptive mapping, plots, and exploratory effect synthesis.	Reported
Critical appraisal	Methods state that formal risk-of-bias appraisal was not the central scoping-review objective; reporting completeness is summarized instead.	Reported
Limitations	Discussion describes database/source-set, extraction, reporting, heterogeneity, and meta-analysis limitations.	Reported
Funding and conflicts	Declarations section included.	Reported

Table 5 provides a reporting crosswalk for the main PRISMA-ScR items. It is included to make the review structure transparent and to show where selection, charting, synthesis, limitations, and declarations are addressed in the manuscript.

B Supplementary Study Characteristics

Table S1: Study-level baseline characteristics of the 65 included reports.

Study	Year	Design	Sport/context	Athlete level	N	Age mean (SD)	Male (%)
Durand2000	2000	Observational	Cycling, Triathlon, Running	elite	22	23.9 (1.3)	100
Spiering2004	2004	Observational	Winter sports	elite	78	NR	32
Lindholm2005	2005	Observational	Breath-hold diving	competitive	5	NR	NR
Woorons2008	2008	RCT	Running	competitive	16	27.1 (6)	NR
Guimard2014	2014	Crossover	Swimming	competitive	15	21.9 (0.9)	100
Malakhov2014	2014	Crossover	Field athletics, team sports	competitive	66	19.8 (1)	44
Sakamoto2014	2014	Crossover	Cycling	competitive	15	NR	NR
Fujii2015	2015	Crossover	Cycling	unclear	11	NR	NR
Jacob2015	2015	Crossover	Swimming	competitive	14	NR	NR
Sakamoto2015	2015	Crossover	Resistance training	competitive	15	21.2 (3.7)	73
Stavrou2015	2015	RCT	Fin-swimming	competitive	28	NR	50
Woorons2016	2016	RCT	Swimming	competitive	16	NR	75
Burtch2017	2017	RCT	Swimming	competitive	25	NR	NR
Hakked2017	2017	RCT	Swimming	competitive	30	NR	NR
Kacoglu2017	2017	Crossover	Jumping and sprinting	healthy active	14	NR	NR
Pelka2017	2017	Crossover	Sprinting	competitive	27	25.22 (1.08)	70
Trincat2017	2017	RCT	Swimming	competitive	16	15.55 (1.8)	56
Woorons2017	2017	Crossover	Cycling	competitive	9	27.2 (9.3)	89
FornasierSantos2018	2018	RCT	Rugby	competitive	35	NR	100
Gray2018	2018	Crossover	Swimming	competitive	7	NR	0
Lim2018	2018	Crossover	Cycling	recreational	10	24 (3)	100
Sakamoto2018	2018	Crossover	Cycling	competitive	17	NR	88

Continued on next page

Table S1: Study-level baseline characteristics of the 65 included reports (continued).

Study	Year	Design	Sport/context	Athlete level	N	Age mean (SD)	Male (%)
Fernandez2019	2019	Observational	Freediving	competitive	25	NR	100
Laborde2019	2019	Crossover	Burpees	healthy active	120	25.57 (2.52)	58
Woorons2019	2019	Crossover	Shuttle run sprints	competitive	10	NR	NR
Woorons2019b	2019	RCT	Cycling	competitive	18	34.6 (11)	100
Bouten2020	2020	Crossover	Cycling	competitive	11	22.1 (1.5)	NR
Lapointe2020	2020	RCT	Basketball	competitive	17	22.3 (1.2)	71
Robertson2020	2020	Crossover	Swimming	competitive	9	19 (1.5)	67
Sakamoto2020	2020	Crossover	Bench press, leg press	competitive	11	22.5 (4.3)	100
Woorons2020	2020	RCT	Team sports	competitive	16	NR	NR
Bahensky2021	2021	RCT	Unclear	unclear	46	NR	NR
Citherlet2021	2021	Crossover	Repeated sprinting	competitive	16	NR	NR
Dobashi2021	2021	Crossover	Cycling	mixed	11	24 (2)	91
Grossman2021	2021	Crossover	Cycling	healthy active	10	23.7 (2.5)	100
Taylor2021	2021	Crossover	Memory-guided force pulse	healthy active	47	21.3 (2.8)	36
Vaidya2021	2021	RCT	Cricket	competitive	30	NR	NR
Brocherie2022	2022	RCT	Ice hockey	elite	43	16.7 (1.5)	100
Conlon2022	2022	RCT	Shooting	competitive	67	NR	NR
Marko2022	2022	Before-after	Cycling	healthy active	23	NR	NR
You2023	2023	Crossover	Unclear	healthy active	78	22.32 (2.11)	71
Braham2024	2024	RCT	Soccer	competitive	16	NR	0
Buxton2024	2024	Crossover	Back squat	recreational	18	20.7 (1.4)	100
Li2024	2024	Crossover	Burpees	healthy active	30	19.6 (1.2)	82

Continued on next page

Table S1: Study-level baseline characteristics of the 65 included reports (continued).

Study	Year	Design	Sport/context	Athlete level	N	Age mean (SD)	Male (%)
Merlin2024	2024	RCT	Swimming	elite	13	13.9 (0.38)	38
Rosa2024	2024	Crossover	Arm cycle ergometer	unclear	14	NR	NR
Sikora2024	2024	Observational	Unclear	competitive	69	NR	NR
Woorons2024	2024	RCT	Judo	elite	20	20.6 (2.5)	50
Deniz2025	2025	Crossover	Bench press	recreational	12	27.92 (7.38)	100
Fernandez2025	2025	Crossover	Freediving	competitive	13	42 (9)	NR
Fessler2025	2025	Crossover	Sprint intervals	recreational	36	24 (1.8)	50
Fox2025	2025	RCT	General	healthy active	404	37 (9.95)	44
Kasap2025	2025	Crossover	Cycle ergometer	healthy active	40	20.95 (1.75)	63
Liapaki2025	2025	Crossover	Running	competitive	15	24.7 (3.1)	NR
Lorinczi2025	2025	RCT	Soccer	competitive	20	NR	NR
Yilmaz2025	2025	Other experimental	Wingate	healthy active	12	NR	NR
Chiron2026	2026	RCT	Running	elite	22	22.3 (3)	68
Devipriya2026	2026	Before-after	Unclear	healthy active	60	21.35 (1.78)	50
FernandezBarradas2026	2026	Before-after	American football	competitive	34	16.97 (0.8)	NR
Iskra2026	2026	Crossover	Unclear	healthy active	58	NR	NR
Jing2026	2026	RCT	Swimming	competitive	20	NR	NR
Jones2026	2026	RCT	Rowing	healthy active	77	21 (3)	71
Katz2026	2026	Crossover	Deadlift	recreational	40	24.8 (6.8)	80
Raidl2026	2026	RCT	Sprint interval training	competitive	20	NR	NR
Tomita2026	2026	Crossover	Speed skating	elite	26	NR	NR

Note. NR = not reported; values are shown as reported or extractable from the source papers.

Supplementary Table S1 provides the study-level records behind the aggregate summaries in the main text. It is placed in the appendix because the 65-report format is useful for transparency but too large for the main Results flow.

C Supplementary Figures

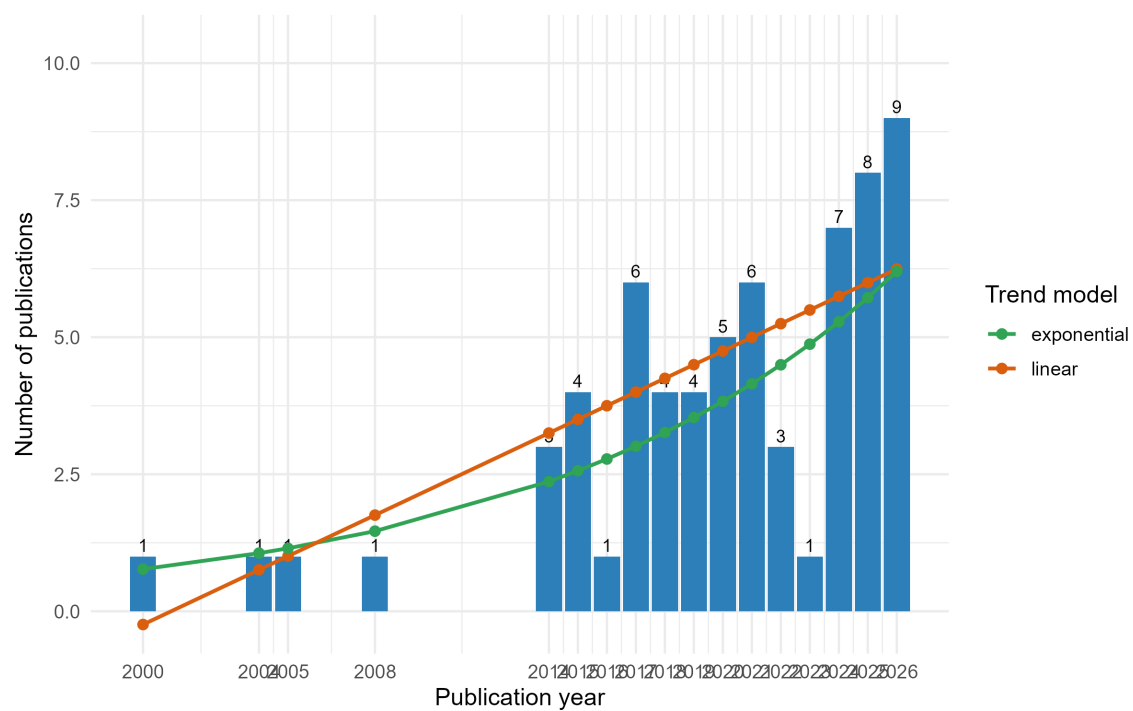


Figure S1: Annual publication counts and fitted trend lines.

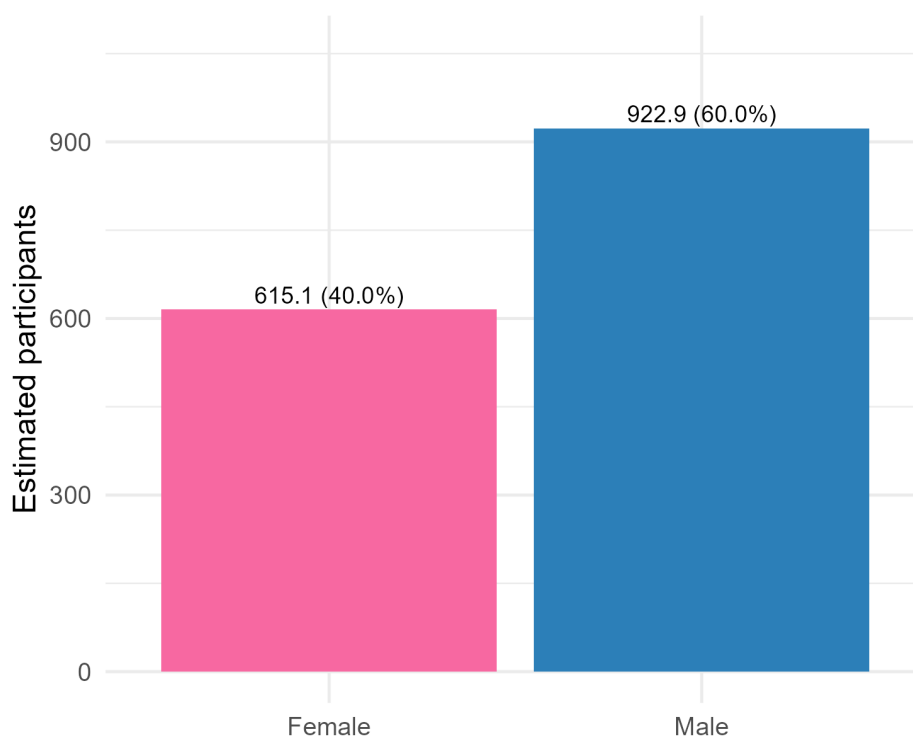


Figure S2: Estimated sex distribution among studies with extractable sex data.

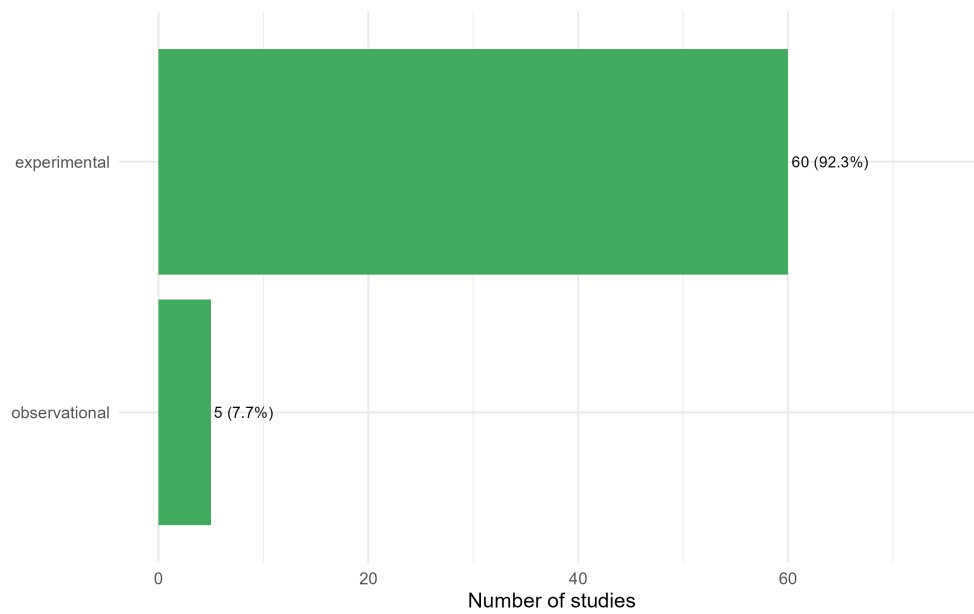


Figure S3: Broad study-design distribution.

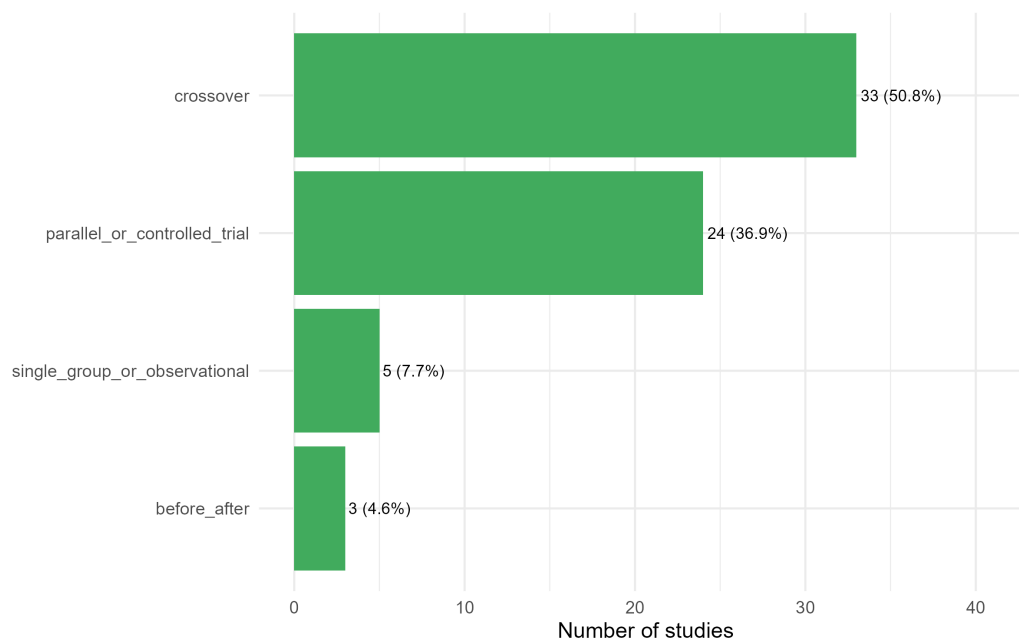


Figure S4: Detailed study-design distribution.

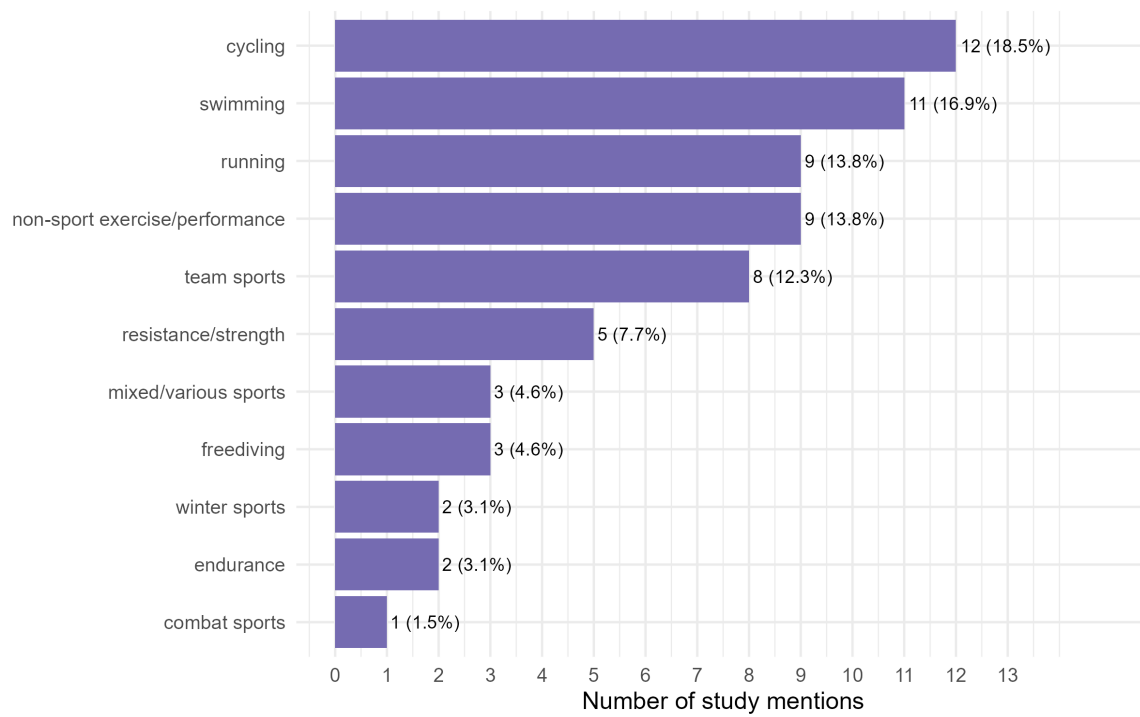


Figure S5: Distribution of sport and exercise contexts.

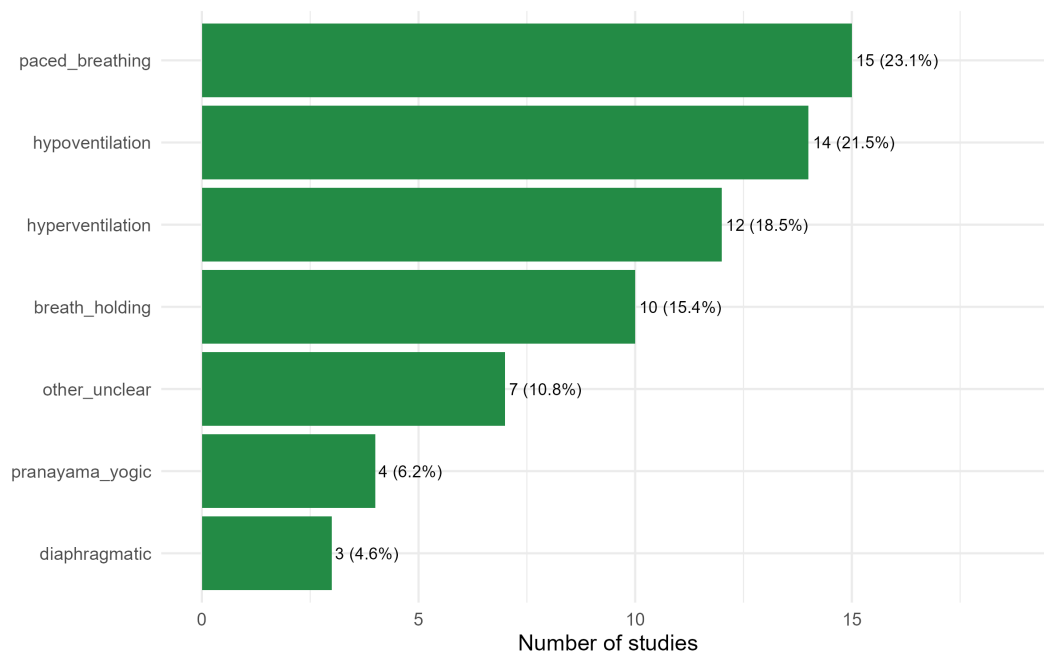


Figure S6: Breathing-category distribution.

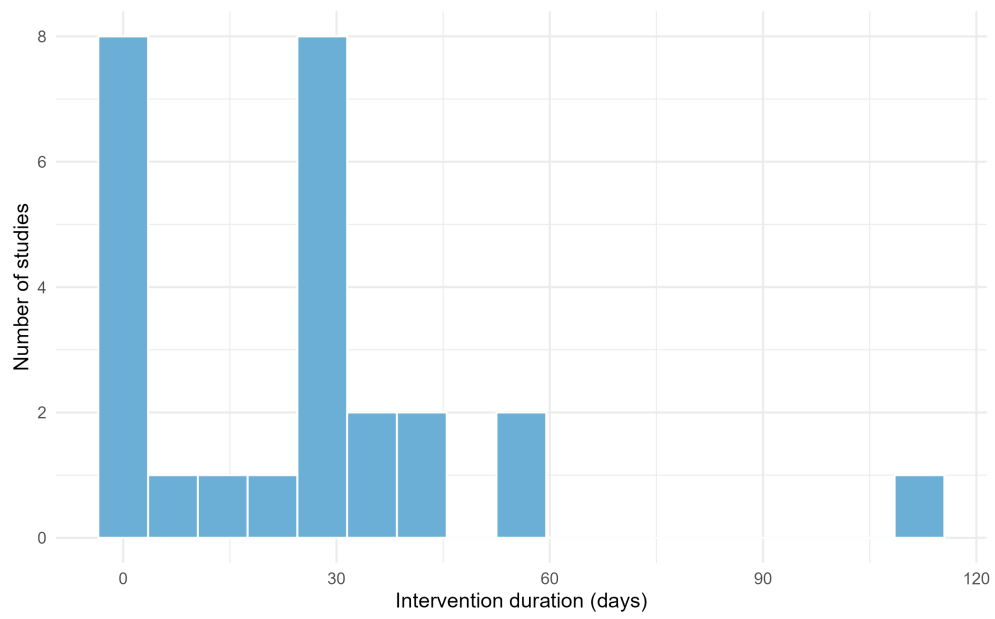


Figure S7: Intervention-duration distribution among reports with extractable duration.

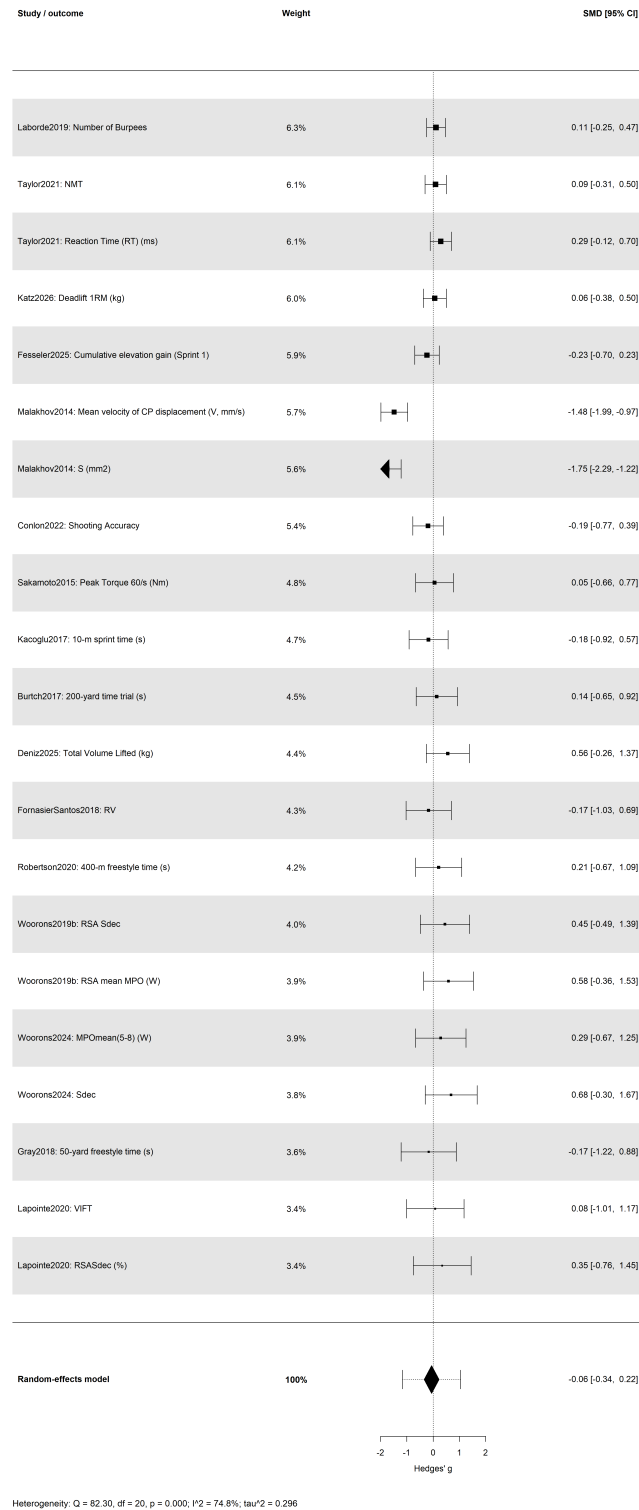


Figure S8: Forest plot for performance outcomes. Effects are standardized so positive Hedges' g favors the breathing intervention.

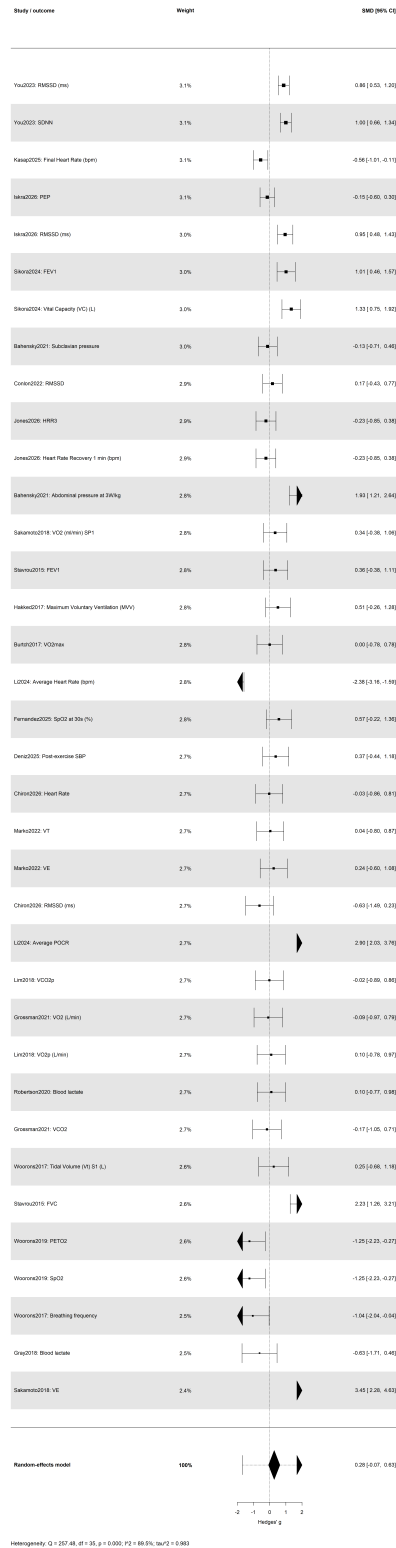


Figure S9: Forest plot for physiological outcomes. Effects are standardized so positive Hedges' g favors the breathing intervention.

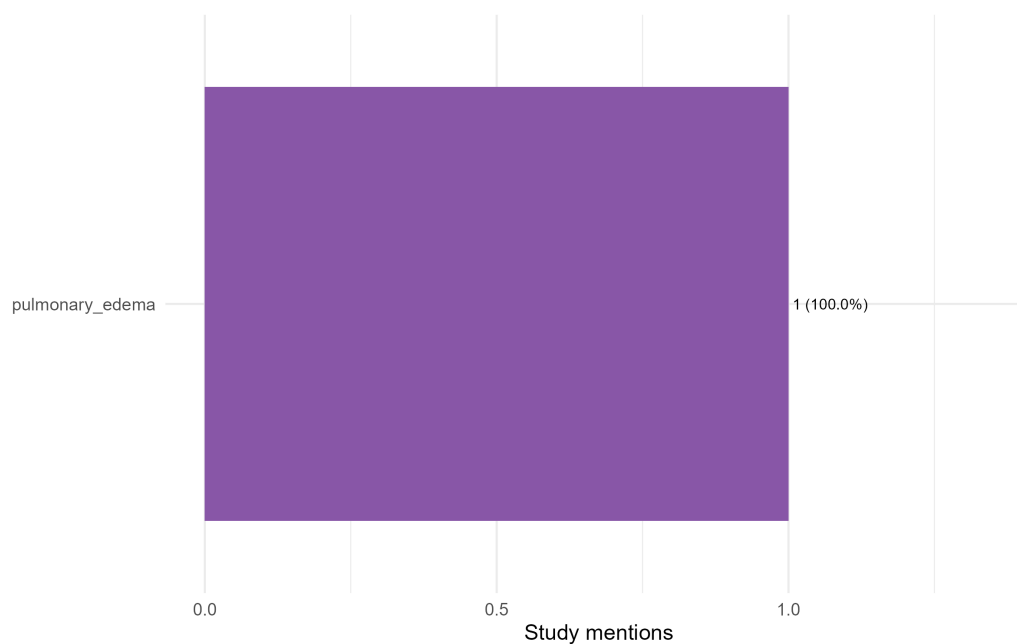


Figure S10: Adverse-event mentions by breathing-technique category.

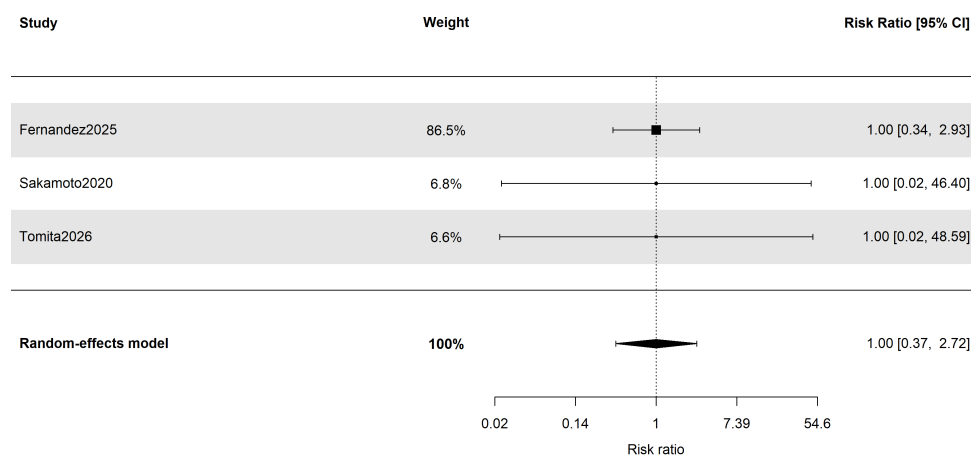


Figure S11: Safety forest plot for adverse events.

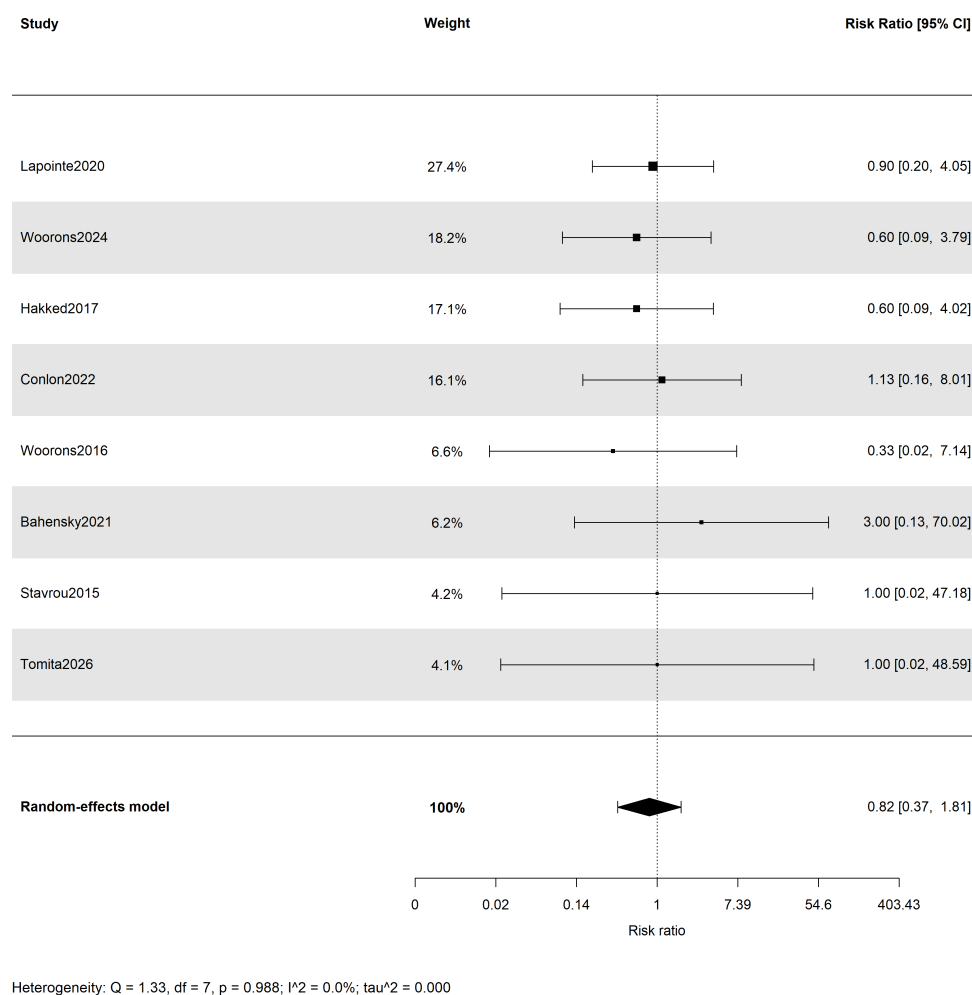


Figure S12: Feasibility forest plot for dropout risk.

D Exact Search Strings

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E Inclusion Criteria

Eligible studies included healthy participants, athletes, exercisers, trained participants, recreationally active participants, sport students, or healthy volunteers. Mixed samples were eligible when data for healthy, athletic, trained, or physically active participants were reported separately, or when the sample was not primarily clinical.

Eligible concepts were breathing interventions, breathing maneuvers, or breathing-related exposures relevant to sport, exercise, or performance. Eligible strategies included breath holding, apnea or apnoea, voluntary hypoventilation, reduced-frequency breathing during exercise, voluntary hyperventilation, hypocapnic hyperventilation, deep or rapid breathing intended to alter ventilation, paced breathing, slow-paced breathing, coherent breathing, resonance breathing, metronome breathing, controlled breathing, diaphragmatic or abdominal breathing, pranayama or yogic breathing, Bhramari breathing, alternate-nostril or uni-nostril breathing, Wim Hof-type breathing, box breathing, hook breathing, and other structured breath-regulation practices. HRV biofeedback or respiratory biofeedback was eligible when the biofeedback was used to guide,

monitor, or train breathing rather than acting as the main physiological intervention.

Eligible contexts included sport, exercise, training, recovery, competition, and performance-relevant laboratory contexts. Eligible outcomes included sport-, exercise-, or physical-performance-relevant outcomes such as endurance, sprint performance, time-trial performance, repeated-sprint ability, power, strength, jump performance, swimming, cycling, running performance, skill accuracy, shooting performance, reaction time, balance, fatigue resistance, recovery performance, or other objective or validated performance-related outcomes. Physiological, psychological, perceptual, feasibility, adherence, or safety outcomes were eligible as secondary outcomes when the study also had a sport-, exercise-, or performance-relevant context.

Original empirical studies of any design were eligible, including randomized controlled trials, crossover studies, non-randomized experimental studies, before-after studies, observational studies, and pilot studies. Studies were not excluded solely because they were small, acute, non-randomized, or exploratory, provided they met the population, concept, context, and outcome criteria.

F Exclusion Criteria

Studies were excluded if they were reviews, systematic reviews, scoping reviews, meta-analyses, narrative reviews, editorials, commentaries, letters without original data, protocols without results, or theoretical papers.

Studies were excluded if they were conducted primarily in clinical or patient populations unless a healthy, athletic, trained, recreationally active, or physically active subgroup was reported separately. Examples of excluded clinical populations included COPD, asthma, sleep apnea, stroke, pulmonary disease, dyspnea-related conditions, cardiac disease, neurological disease, and rehabilitation-only samples.

Studies were excluded if they did not test or observe a breathing maneuver, breathing intervention, or breath-regulation strategy. Respiratory muscle training, inspiratory muscle training, expiratory muscle training, respiratory loading, resisted breathing, pressure-threshold loading, aided inspiration, oxygen supplementation, hypoxic chambers, altitude tents, altered gas mixtures, or other device-based interventions were excluded when the device or altered gas condition was the main intervention rather than self-regulated breathing. Meditation, relaxation, mindfulness, yoga, or biofeedback interventions were excluded when the breathing component was absent, not described, or could not be separated from a broader multicomponent intervention.

Studies were excluded if they had no sport, exercise, physical-performance, recovery, competition, or performance-relevant context, or if they reported only physiological, psychological, autonomic, biochemical, or perceptual outcomes without a performance-relevant task or outcome. Abstracts without sufficient data, unavailable full texts, duplicate records, animal studies,

non-human studies, and non-English articles were also excluded when applicable.

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